

MACE HEAD, Ireland

WMO: 039630

Lat: 53.3267N

Lon: 9.9043W

Elev: 23

StdP: 101.05

Time Zone: 1.00 (IRL)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.8	0.7	-4.7	2.5	0.5	-2.9	3.0	1.9	22.0	7.1	20.2	7.1	4.8	50	0.825

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	20.6	16.7	18.9	15.7	17.7	15.1	17.4	19.6	16.5	18.1	15.9	17.1	5.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	11.8	18.2	15.8	11.3	17.2	15.3	10.9	16.7	48.9	19.6	46.4	18.3	44.5	17.2	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.4	14.7	DB	-1.8	24.0	2.2	2.7	-3.3	25.9	-4.6	27.5	-5.8	29.0	-7.4	30.9	(4)
(5)			WB	-2.8	18.7	2.3	1.5	-4.5	19.7	-5.8	20.6	-7.1	21.5	-8.8	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	6.7	6.5	7.3	9.3	11.5	13.7	15.1	15.0	13.9	11.6	8.8	7.4		
	DBStd	3.79	2.42	2.24	2.31	2.00	2.21	2.01	1.71	1.16	1.56	1.98	2.37	2.75		
	HDD10.0	468	102	98	86	35	7	0	0	0	0	8	49	83		
	HDD18.3	2833	361	331	341	271	213	141	103	104	133	210	287	338		
	CDD10.0	683	1	1	3	14	52	112	158	154	117	57	12	3		
	CDD18.3	5	0	0	0	0	1	2	2	1	0	0	0	0		
	CDH23.3	14	0	0	0	0	2	8	3	1	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0		
(14) Wind	WSAvg	7.5	8.5	8.2	7.3	6.7	6.9	6.1	6.1	7.2	7.5	7.9	8.2	9.2		
(15) Precipitation	PrecAvg	1666	178	141	118	102	99	102	113	130	131	165	187	188		
	PrecMax	1929	305	273	219	226	185	177	171	289	241	320	337	331		
	PrecMin	1294	61	40	58	46	29	52	56	36	23	56	92	78		
	PrecStd	177	59	62	38	38	44	36	33	59	59	66	55	72		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	11.7	15.2	17.9	22.1	24.1	23.3	20.5	19.7	17.3	14.1	12.6		
		MCWB	10.6	10.1	10.5	12.7	16.7	17.3	18.9	16.7	15.9	15.3	13.1	11.6		
	2%	DB	10.6	10.7	12.2	14.8	18.5	20.1	20.5	18.4	17.9	15.7	13.3	12.0		
		MCWB	9.9	9.7	9.4	10.8	14.3	15.8	17.5	15.7	15.4	14.4	12.3	11.2		
	5%	DB	10.2	10.1	10.8	13.2	16.5	18.1	18.7	17.4	16.7	14.7	12.6	11.3		
		MCWB	9.5	9.4	9.0	10.4	13.1	14.7	16.4	15.1	15.0	13.3	11.6	10.6		
	10%	DB	9.9	9.6	10.1	12.1	14.7	16.7	17.6	16.7	16.0	14.1	12.0	10.8		
		MCWB	9.1	8.8	8.8	9.7	11.9	14.0	15.4	14.7	14.4	12.9	10.9	10.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.6	11.1	13.2	16.8	17.9	19.2	17.5	16.9	15.9	13.5	12.2		
		MCDB	11.3	11.1	13.5	17.1	21.6	22.8	22.5	19.8	17.8	16.6	13.8	12.3		
	2%	WB	10.3	10.0	10.1	11.6	14.8	16.3	17.9	16.5	16.2	14.9	12.6	11.5		
		MCDB	10.6	10.5	11.3	13.3	18.1	19.8	19.9	17.6	17.1	15.6	13.1	11.8		
	5%	WB	9.9	9.6	9.7	10.9	13.6	15.2	16.9	15.9	15.5	13.9	12.0	10.7		
		MCDB	10.2	10.1	10.6	12.4	15.8	17.4	18.4	16.9	16.4	14.4	12.5	11.2		
	10%	WB	9.4	9.1	9.1	10.4	12.6	14.5	16.0	15.4	15.0	13.3	11.2	10.3		
		MCDB	9.9	9.7	10.1	11.9	14.2	16.2	17.1	16.4	15.8	14.0	12.0	10.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.8	4.3	5.1	4.9	4.6	3.9	3.6	3.7	3.9	3.7	3.4		
		MCDBR	3.0	3.2	4.9	6.5	8.0	7.4	5.5	4.4	4.5	3.9	3.5	3.3		
	5% WB	MCWBR	3.5	3.2	3.7	4.1	4.8	4.2	3.6	3.1	3.5	3.6	3.8	3.7		
		MCWBR	2.8	3.1	4.4	5.4	6.9	6.0	5.0	3.8	3.8	3.2	3.6	3.3		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.344	0.359	0.376	0.385	0.368	0.373	0.361	0.361	0.343	0.332	0.326		
		taud	2.276	2.365	2.380	2.356	2.373	2.458	2.453	2.498	2.481	2.471	2.329	2.243		
	Ebn at Noon		615	753	824	855	864	883	871	864	816	749	620	544		
		Edh at Noon	63	79	94	109	113	105	104	94	85	70	59	54		
(44) All-Sky Solar Radiation	RadAvg		0.66	1.35	2.52	4.04	5.03	5.28	4.68	3.99	2.88	1.68	0.82	0.49		
	RadStd		0.06	0.12	0.22	0.34	0.39	0.46	0.34	0.31	0.21	0.13	0.07	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page